

MoodTune

Facial Expression Recognition
& Music Recommendation via Deep Learning

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MT MoodTune

WHAT I SEE

Angry

Mood group: stressed

60
%

MT MoodTune

WHAT I SEE

Happy

Mood group: positive

87
%

Agenda

01 Problem Statement

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03 System Architecture

04 CNN Model Design

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Problem Statement

Emotional overwhelm

Choosing music while overwhelmed is cognitively demanding — it's hardest exactly when you need it most.

Manual input bottleneck

Existing systems require users to self-report mood — imprecise, slow, and breaks the emotional flow.

Generic playlists

Streaming platforms use listening history, not current emotion — missing real-time affective context entirely.

Untapped signal

Facial expressions are the fastest, richest signal for visible mood — yet no mainstream app leverages them live.

OUR SOLUTION — MOODTUNE

Scan face → detect expression → map to mood → stream perfectly matched music. Zero manual input required.

RAF-DB Dataset

12,271

Training images

3,068

Test images

7

Emotion classes

CLASS DISTRIBUTION — TRAINING SET

Happiness



Neutral

2,524

Sadness

1,982

Surprise

1,290

Disgust

717

Anger

705

Fear

281

WHY RAF-DB OVER FER2013?

Resolution

Color · High-res

Label quality

Cleaner [crowdsourced](#)

Acc. ceiling

85–88%

FER2013 noise

~20% [label error](#)

CLASS IMBALANCE HANDLING

Inverse-frequency class weights applied during training:

Fear

$w = 6.23\times$

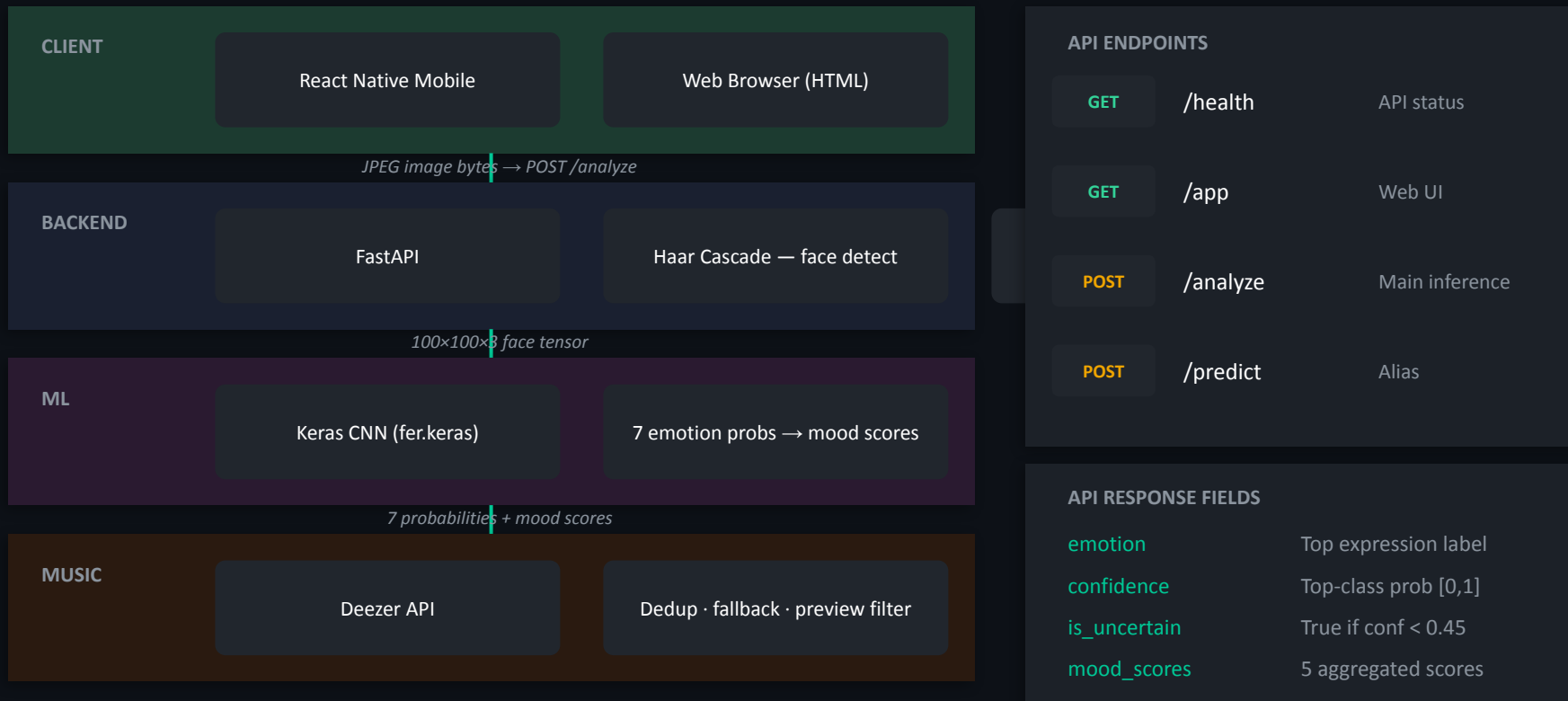
Disgust

$w = 2.50\times$

Happiness

$w = 0.37\times$

System Architecture



CNN Architecture

LAYER-BY-LAYER FLOW

Input · 100×100×3	
Conv2D 64 · 3×3 · ReLU	1,792 params
MaxPool 2×2 → 49×49×64	
Conv2D 64 · 3×3 · ReLU	36,928 params
MaxPool 2×2 → 23×23×64	
Conv2D 128 · 3×3 · ReLU	73,856 params
MaxPool 2×2 → 10×10×128	
Flatten → 12,800-dim vector	
Dense 256 · ReLU + Dropout	3,277,056 params
Dense 7 · Softmax	1,799 params

3.4M

TOTAL PARAMETERS

96.6%

PARAMS IN DENSE 256

TRAINING CONFIG

Optimizer	Adam · lr=0.001
Loss	Sparse cat. cross-entropy
Batch size	32 → 384 steps/epoch
Max epochs	50 + EarlyStopping
Val split	25% of training set
Callbacks	ModelCheckpoint, ReduceLR

Experimental Results

76.60%

TEST ACCURACY

0.7619

WEIGHTED F1

0.6601

TEST LOSS

3,068

TEST SAMPLES

F1-SCORE PER CLASS



WHY THE GAP BETWEEN CLASSES?

Happiness (F1=0.89)

Largest class (4,772 samples) + visually distinctive symmetric smile.
Easiest to learn.

Surprise (F1=0.78)

Distinctive wide-eye + brow raise. Unique spatial pattern.

Fear (F1=0.50)

Only 281 samples. Shares wide-eye features with Surprise. Hard boundary.

Disgust (F1=0.36)

Subtle nose wrinkle overlaps with Sadness in mouth region. Most ambiguous visually.

💡 Mood aggregation compensates: Fear + Disgust + Anger all map to Stressed — confusion within this group still produces a correct recommendation.

Mood Aggregation Logic

EMOTION → MOOD GROUP MAPPING

 **Stressed**

Anger

Fear

Disgust

 **Positive**

Happiness

 **Low**

Sadness

 **Energetic**

Surprise

 **Calm**

Neutral

Why not just pick the top class?

When predictions are close — e.g. Fear 24%, Disgust 24%, Anger 21% — a single top-class pick ignores 69% of the signal. Aggregation captures the full probability mass.

EXAMPLE: AMBIGUOUS STRESSED STATE

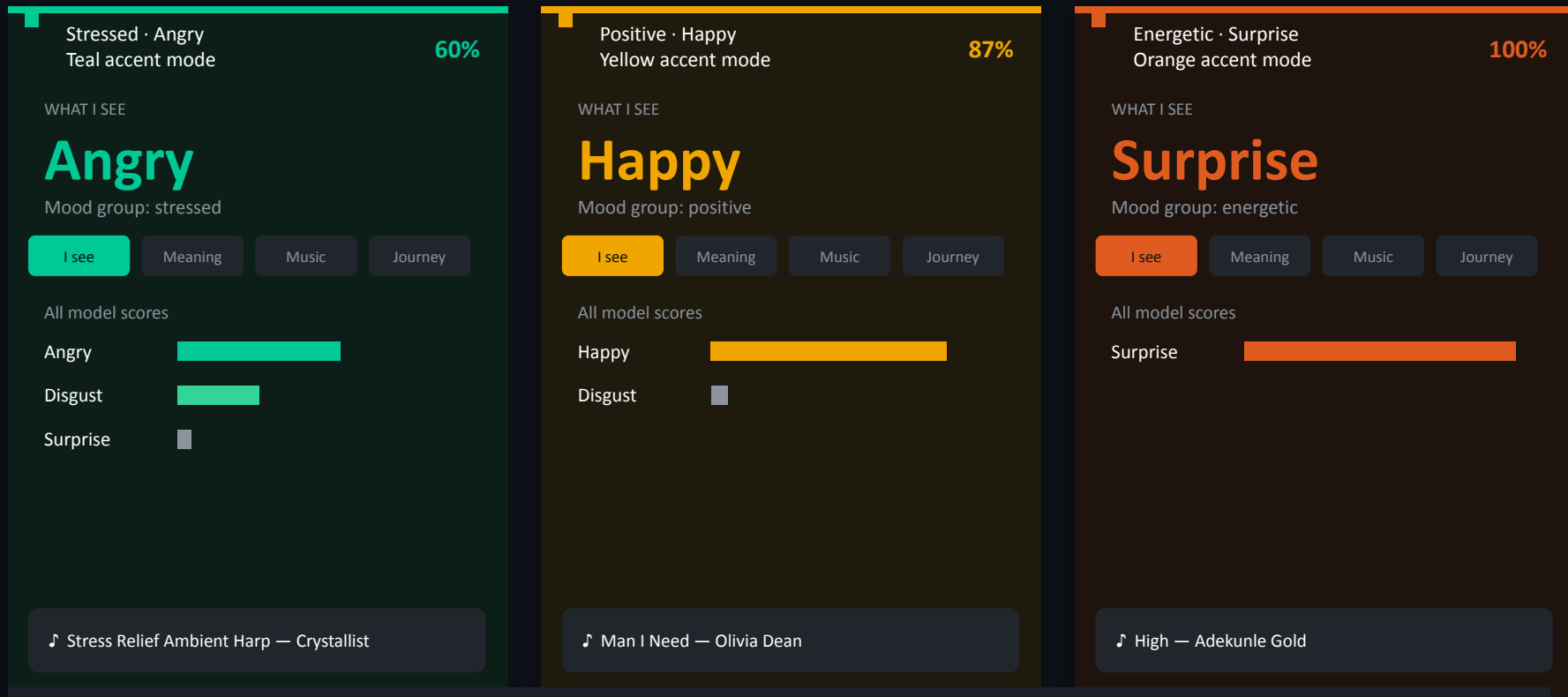
Fear		24%
Disgust		24%
Anger		21%

Stressed score = 24 + 24 + 21 = 69% → dominant mood ✓

Uncertainty handling

Predictions below 0.45 confidence are flagged as `is_uncertain: true` — never silently forced to a default label. Transparency over false confidence.

User Interface — Live App



DYNAMIC UI Accent color, ring, tabs, coaching, and music track all adapt to detected mood in real time. Zero manual input.

Conclusion & Future Work

ACHIEVEMENTS

Custom CNN from scratch

3.4M params · trained on 12,271 RAF-DB images · 76.60% accuracy · Weighted F1 0.76



FastAPI + Deezer integration

Clean ML/UI boundary · lazy model loading · real preview tracks for every mood

Mood aggregation system

5 mood groups · probability-weighted · uncertainty flagging · honest confidence reporting

Dual interface

Expo React Native mobile + browser web app · dynamic mood-adaptive UI

FUTURE WORK

01

User correction button — "Does this mood feel right?" ground truth feedback loop

02

Fine-tune on phone-camera images — close the lab/real-world distribution gap

03

Oversample Fear & Disgust — synthetic augmentation to boost hardest classes

04

TensorFlow Lite conversion — on-device inference, no server, better privacy

05

Multimodal signals — voice tone + self-reported mood for richer context

06

Anonymised feedback loop — personalise recommendations over time